TBL Session: Hypo- and Hyperglycemia

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Block: Endocrine/Reproduction/Urology
Thread: Biochemistry
Conflict of Interest: Dr. Scott has no conflicts of interest to disclose.

Pre-Work

You should be familiar with the basic content of the following lessons:
BCM Regulation of Plasma Glucose (October 6th, 2026)
BCM Dysregulation of Metabolism in Diabetes (October 6th, 2026)

Learning Objectives for Teamwork

On completing this TBL session and as you prepare for the end-of-block exam, you should be able to do the following:

1. Recognize the symptoms and abnormalities in laboratory values in persons who present with diabetic ketoacidosis versus hyperosmolar hyperglycemic state.
2. For diabetic ketoacidosis, compare and contrast abnormalities of flux in metabolic pathways that normally play a role in energy production.
3. Provide a threshold concentration for glucose and for ketone bodies in blood at which these substances appear in urine.
4. Describe the effects of hypophosphatemia on ATP production and the affinity of hemoglobin for oxygen.
5. Recognize the symptoms and abnormalities in laboratory values in persons who present with hypoglycemia due to glycogen storage disease type 1 (von Gierke disease).
6. For glycogen storage disease type 1 (von Gierke disease), describe and interpret abnormalities of flux in glycogen metabolism, gluconeogenesis, triglyceride metabolism, fatty acid oxidation, and ketone body metabolism. Then, explain how treatment can minimize these abnormalities.